

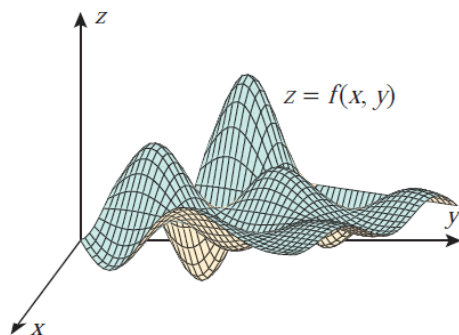
13.8 MAXIMA AND MINIMA OF FUNCTIONS OF TWO VARIABLES

We know how to find maximum and minimum values of a function of one variable. In this section we will develop similar techniques for functions of two variables.

■ EXTREMA

If we imagine the graph of a function f of two variables to be a mountain range (Figure 13.8.1), then the mountaintops, which are the high points in their immediate vicinity, are called *relative maxima* of f , and the valley bottoms, which are the low points in their immediate vicinity, are called *relative minima* of f .

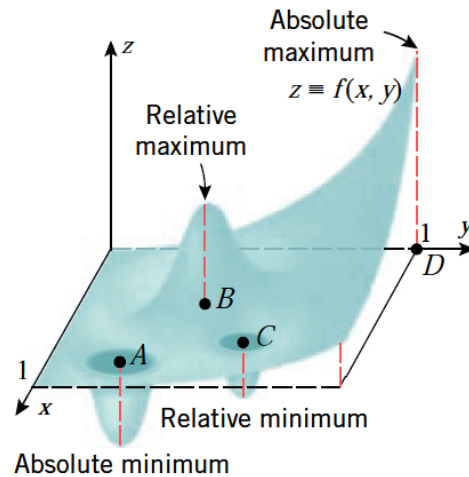
Just as a geologist might be interested in finding the highest mountain and deepest valley in an entire mountain range, so a mathematician might be interested in finding the largest and smallest values of $f(x, y)$ over the *entire* domain of f . These are called the *absolute maximum* and *absolute minimum values* of f . The following definitions make these informal ideas precise.



▲ Figure 13.8.1

13.8.1 DEFINITION A function f of two variables is said to have a ***relative maximum*** at a point (x_0, y_0) if there is a disk centered at (x_0, y_0) such that $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) that lie inside the disk, and f is said to have an ***absolute maximum*** at (x_0, y_0) if $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the domain of f .

13.8.2 DEFINITION A function f of two variables is said to have a ***relative minimum*** at a point (x_0, y_0) if there is a disk centered at (x_0, y_0) such that $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) that lie inside the disk, and f is said to have an ***absolute minimum*** at (x_0, y_0) if $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) in the domain of f .



▲ Figure 13.8.2

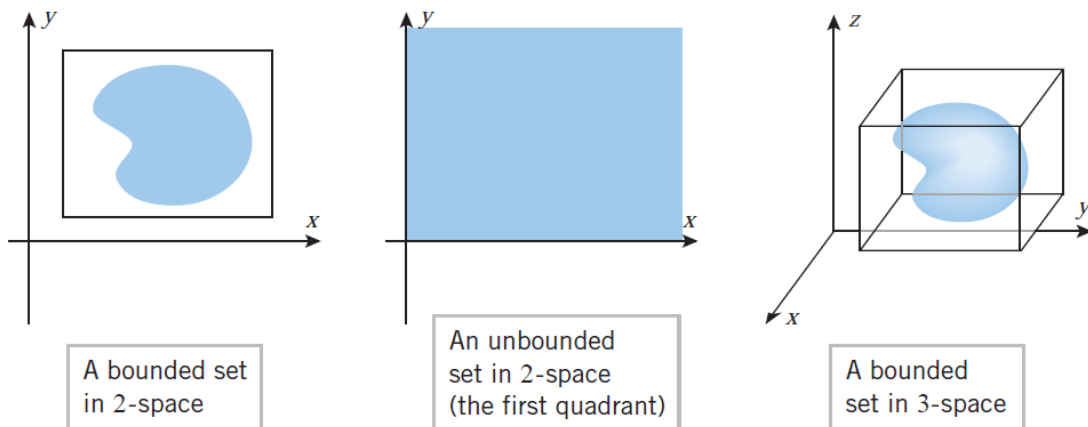
If f has a relative maximum or a relative minimum at (x_0, y_0) , then we say that f has a **relative extremum** at (x_0, y_0) , and if f has an absolute maximum or absolute minimum at (x_0, y_0) , then we say that f has an **absolute extremum** at (x_0, y_0) .

Remark-1:

For functions of two variables we will be concerned with two important questions:

- Are there any relative or absolute extrema?
- If so, where are they located?

Note for Bounded Set:



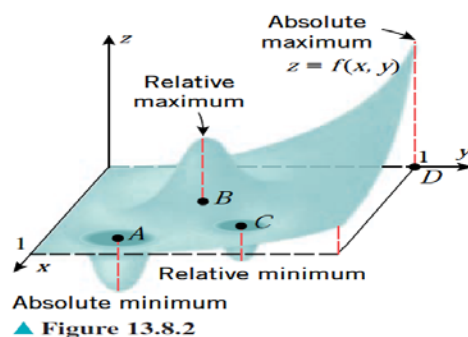
The Extreme-value Theorem:

13.8.3 THEOREM (Extreme-Value Theorem) If $f(x, y)$ is continuous on a closed and bounded set R , then f has both an absolute maximum and an absolute minimum on R .

► **Example 1** The square region R whose points satisfy the inequalities

$$0 \leq x \leq 1 \quad \text{and} \quad 0 \leq y \leq 1$$

is a closed and bounded set in the xy -plane. The function f whose graph is shown in Figure 13.8.2 is continuous on R ; thus, it is guaranteed to have an absolute maximum and minimum on R by the last theorem. These occur at points D and A that are shown in the figure. ◀



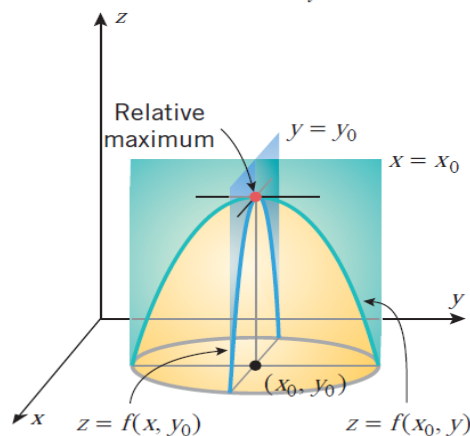
Remark-2:

If any of the conditions in the Extreme-Value Theorem fail to hold, then there is no guarantee that an absolute maximum or absolute minimum exists on the region R . Thus, a discontinuous function on a closed and bounded set need not have any absolute extrema, and a continuous function on a set that is not closed and bounded also need not have any absolute extrema.

Note on Finding Relative Extrema:

13.8.4 THEOREM If f has a relative extremum at a point (x_0, y_0) , and if the first-order partial derivatives of f exist at this point, then

$$f_x(x_0, y_0) = 0 \quad \text{and} \quad f_y(x_0, y_0) = 0$$



Recall that the *critical points* of a function f of one variable are those values of x in the domain of f at which $f'(x) = 0$ or f is not differentiable. The following definition is the analog for functions of two variables.

13.8.5 DEFINITION A point (x_0, y_0) in the domain of a function $f(x, y)$ is called a **critical point** of the function if $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$ or if one or both partial derivatives do not exist at (x_0, y_0) .

Note-1: Critical point, point of inflection and saddle node

Recall that for a function of one variable a relative extremum need not occur at *every* critical point. For example, the function might have an inflection point with a horizontal tangent line at the critical point. Similarly, a function of two variables need not have a relative extremum at every critical point. For example, consider the function

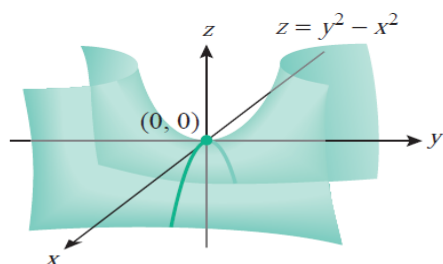
$$f(x, y) = y^2 - x^2$$

This function, whose graph is the hyperbolic paraboloid shown in Figure 13.8.5, has a critical point at $(0, 0)$, since

$$f_x(x, y) = -2x \quad \text{and} \quad f_y(x, y) = 2y$$

from which it follows that

$$f_x(0, 0) = 0 \quad \text{and} \quad f_y(0, 0) = 0$$

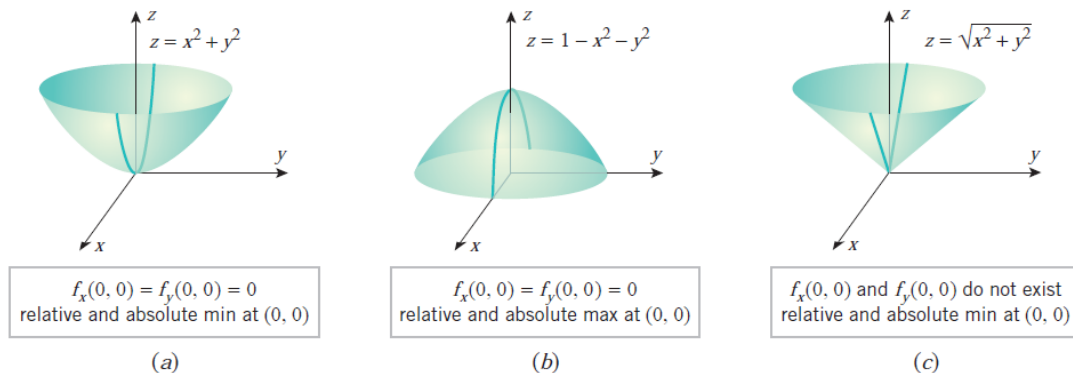


The function $f(x, y) = y^2 - x^2$ has neither a relative maximum nor a relative minimum at the critical point $(0, 0)$.

▲ Figure 13.8.5

However, the function f has neither a relative maximum nor a relative minimum at $(0, 0)$. For obvious reasons, the point $(0, 0)$ is called a *saddle point* of f . In general, we will say that a surface $z = f(x, y)$ has a **saddle point** at (x_0, y_0) if there are two distinct vertical planes through this point such that the trace of the surface in one of the planes has a relative maximum at (x_0, y_0) and the trace in the other has a relative minimum at (x_0, y_0) .

Observation-1:



▲ Figure 13.8.6

THE SECOND PARTIALS TEST

13.8.6 THEOREM (The Second Partials Test) Let f be a function of two variables with continuous second-order partial derivatives in some disk centered at a critical point (x_0, y_0) , and let

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

- (a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a relative minimum at (x_0, y_0) .
- (b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a relative maximum at (x_0, y_0) .
- (c) If $D < 0$, then f has a saddle point at (x_0, y_0) .
- (d) If $D = 0$, then no conclusion can be drawn.

► **Example 3** Locate all relative extrema and saddle points of

$$f(x, y) = 3x^2 - 2xy + y^2 - 8y$$

Solution. Since $f_x(x, y) = 6x - 2y$ and $f_y(x, y) = -2x + 2y - 8$, the critical points of f satisfy the equations

$$\begin{aligned} 6x - 2y &= 0 \\ -2x + 2y - 8 &= 0 \end{aligned}$$

Solving these for x and y yields $x = 2$, $y = 6$ (verify), so $(2, 6)$ is the only critical point. To apply Theorem 13.8.6 we need the second-order partial derivatives

$$f_{xx}(x, y) = 6, \quad f_{yy}(x, y) = 2, \quad f_{xy}(x, y) = -2$$

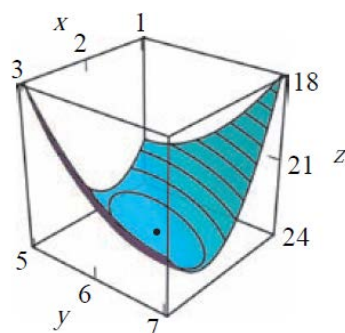
At the point $(2, 6)$ we have

$$D = f_{xx}(2, 6)f_{yy}(2, 6) - f_{xy}^2(2, 6) = (6)(2) - (-2)^2 = 8 > 0$$

and

$$f_{xx}(2, 6) = 6 > 0$$

so f has a relative minimum at $(2, 6)$ by part (a) of the second partials test. Figure 13.8.7 shows a graph of f in the vicinity of the relative minimum. ◀



$$f(x, y) = 3x^2 - 2xy + y^2 - 8y$$

▲ Figure 13.8.7

► **Example 4** Locate all relative extrema and saddle points of

$$f(x, y) = 4xy - x^4 - y^4$$

Solution. Since

$$\begin{aligned} f_x(x, y) &= 4y - 4x^3 \\ f_y(x, y) &= 4x - 4y^3 \end{aligned} \tag{1}$$

the critical points of f have coordinates satisfying the equations

$$\begin{aligned} 4y - 4x^3 &= 0 & \text{or} & & y &= x^3 \\ 4x - 4y^3 &= 0 & & & x &= y^3 \end{aligned} \tag{2}$$

Substituting the top equation in the bottom yields $x = (x^3)^3$ or, equivalently, $x^9 - x = 0$ or $x(x^8 - 1) = 0$, which has solutions $x = 0, x = 1, x = -1$. Substituting these values in the top equation of (2), we obtain the corresponding y -values $y = 0, y = 1, y = -1$. Thus, the critical points of f are $(0, 0)$, $(1, 1)$, and $(-1, -1)$.

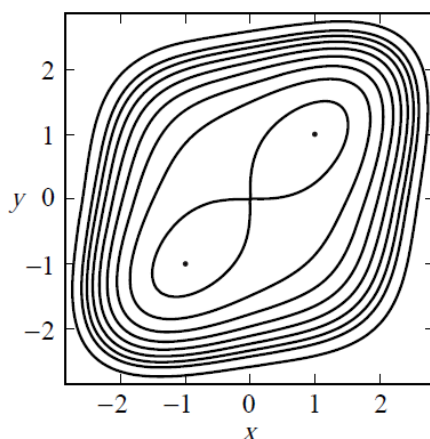
From (1),

$$f_{xx}(x, y) = -12x^2, \quad f_{yy}(x, y) = -12y^2, \quad f_{xy}(x, y) = 4$$

which yields the following table:

CRITICAL POINT (x_0, y_0)	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$	$D = f_{xx}f_{yy} - f_{xy}^2$
(0, 0)	0	0	4	-16
(1, 1)	-12	-12	4	128
(-1, -1)	-12	-12	4	128

At the points (1, 1) and (-1, -1), we have $D > 0$ and $f_{xx} < 0$, so relative maxima occur at these critical points. At (0, 0) there is a saddle point since $D < 0$. The surface and a contour plot are shown in Figure 13.8.8. ◀



▲ Figure 13.8.8

13.8.7 THEOREM *If a function f of two variables has an absolute extremum (either an absolute maximum or an absolute minimum) at an interior point of its domain, then this extremum occurs at a critical point.*

How to Find the Absolute Extrema of a Continuous Function f of Two Variables on a Closed and Bounded Set R

Step 1. Find the critical points of f that lie in the interior of R .

Step 2. Find all boundary points at which the absolute extrema can occur.

Step 3. Evaluate $f(x, y)$ at the points obtained in the preceding steps. The largest of these values is the absolute maximum and the smallest the absolute minimum.

► **Example 5** Find the absolute maximum and minimum values of

$$f(x, y) = 3xy - 6x - 3y + 7 \quad (3)$$

on the closed triangular region R with vertices $(0, 0)$, $(3, 0)$, and $(0, 5)$.

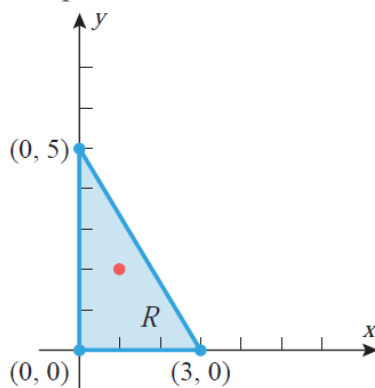
Solution. The region R is shown in Figure 13.8.9. We have

$$\frac{\partial f}{\partial x} = 3y - 6 \quad \text{and} \quad \frac{\partial f}{\partial y} = 3x - 3$$

so all critical points occur where

$$3y - 6 = 0 \quad \text{and} \quad 3x - 3 = 0$$

Solving these equations yields $x = 1$ and $y = 2$, so $(1, 2)$ is the only critical point. As shown in Figure 13.8.9, this critical point is in the interior of R .



▲ Figure 13.8.9

Next we want to determine the locations of the points on the boundary of R at which the absolute extrema might occur. The boundary of R consists of three line segments, each of which we will treat separately:

The line segment between $(0, 0)$ and $(3, 0)$: On this line segment we have $y = 0$, so (3) simplifies to a function of the single variable x ,

$$u(x) = f(x, 0) = -6x + 7, \quad 0 \leq x \leq 3$$

This function has no critical points because $u'(x) = -6$ is nonzero for all x . Thus the extreme values of $u(x)$ occur at the endpoints $x = 0$ and $x = 3$, which correspond to the points $(0, 0)$ and $(3, 0)$ of R .

The line segment between (0, 0) and (0, 5): On this line segment we have $x = 0$, so (3) simplifies to a function of the single variable y ,

$$v(y) = f(0, y) = -3y + 7, \quad 0 \leq y \leq 5$$

This function has no critical points because $v'(y) = -3$ is nonzero for all y . Thus, the extreme values of $v(y)$ occur at the endpoints $y = 0$ and $y = 5$, which correspond to the points (0, 0) and (0, 5) of R .

The line segment between (3, 0) and (0, 5): In the xy -plane, an equation for this line segment is

$$y = -\frac{5}{3}x + 5, \quad 0 \leq x \leq 3 \quad (4)$$

so (3) simplifies to a function of the single variable x ,

$$\begin{aligned} w(x) &= f\left(x, -\frac{5}{3}x + 5\right) = 3x\left(-\frac{5}{3}x + 5\right) - 6x - 3\left(-\frac{5}{3}x + 5\right) + 7 \\ &= -5x^2 + 14x - 8, \quad 0 \leq x \leq 3 \end{aligned}$$

Since $w'(x) = -10x + 14$, the equation $w'(x) = 0$ yields $x = \frac{7}{5}$ as the only critical point of w . Thus, the extreme values of w occur either at the critical point $x = \frac{7}{5}$ or at the endpoints $x = 0$ and $x = 3$. The endpoints correspond to the points (0, 5) and (3, 0) of R , and from (4) the critical point corresponds to $\left(\frac{7}{5}, \frac{8}{3}\right)$.

Finally, Table 13.8.1 lists the values of $f(x, y)$ at the interior critical point and at the points on the boundary where an absolute extremum can occur. From the table we conclude that the absolute maximum value of f is $f(0, 0) = 7$ and the absolute minimum value is $f(3, 0) = -11$. ◀

Table 13.8.1

(x, y)	(0, 0)	(3, 0)	(0, 5)	$\left(\frac{7}{5}, \frac{8}{3}\right)$	(1, 2)
$f(x, y)$	7	-11	-8	$\frac{9}{5}$	1

Practice Problem:

Chapter-13.8:

Problems: 9, 11, 13, 14-19, 42-44

Thank you for your attention!